1. Develop a program to create a structure of library to read and print details of 3 books viz. book ID, title, author and price.

```
#include<conio.h>
#include<stdio.h>

void main()
{
  struct book
  {
    int id;
    char title[30], author[40];
    float price;
  };
  struct book b1, b2, b3;

  printf("Enter book id, title, auther and price \n");
  scanf("%d %s %s %f", &b1.id, b1.title, b1.author, &b1.price);
  scanf("%d %s %s %f", &b2.id, b2.title, b2.author, &b2.price);
  scanf("%d %s %s %f", &b3.id, b3.title, b3.author, &b3.price);

  printf("\nbook id\ttitle \t auther \t price \n");
  printf("%d\t%s\t%s\t%f\n", b1.id, b1.title, b1.author, b1.price);
  printf("%d\t%s\t%s\t%f\n", b2.id, b2.title, b2.author, b2.price);
  printf("%d\t%s\t%s\t%f\n", b3.id, b3.title, b3.author, b3.price);
}
 getch();
}
```

2. Write a program to exchange the values of two variables using call by value method.

```
//Call by Value Example - Swapping 2 numbers using Call by Value
#include <stdio.h>
#include <conio.h>

void swap(int, int);
void main()
{
  int x, y;
 x=10,y=20;
printf("Before Swapping\nx = %d\ny = %d\n", x, y);
swap(x, y);
printf("After Swapping\nx = %d\ny = %d\n", x, y);
getch();
}
void swap(int a, int b)
{
  int temp;
  temp = a;
  a = b;
  b = temp;
   printf("Values of a and b is %d  %d\n",a,b);
}
```

3. Write a program to exchange the values of two variables using call by reference method.

```
//Call by Reference Example - Swapping 2 numbers using Call by Reference
#include <stdio.h>
 #include <conio.h>
 void swap(int*, int*);

void main()
{
   int x, y;
x=10,y=20;
 printf("Before Swapping\nx = %d\ny = %d\n", x, y);
 swap(&x, &y);
 printf("After Swapping\nx = %d\ny = %d\n", x, y);
getch();
}

void swap(int *a, int *b)
{
   int temp;
 temp = *b;
   *b = *a;
   *a = temp;
}
```

4. Write a program to print factorial of number using recursion.

```
#include<conio.h>
#include<stdio.h>
int factorial(int);
void main()
{
  int number;
  long fact;
  printf("Enter a number: ");
  scanf("%d", &number);

  fact = factorial(number);
  printf("Factorial of %d is %ld\n", number, fact);
 getch();
}
int factorial(int n)
{
  if (n == 0)
    return 1;
  else
    return(n * factorial(n-1));
}
```

5. Write a program to print Fibonacci series

```
#include<conio.h>
#include<stdio.h>

void main()

{
 int n1=0,n2=1,n3,i,number;
 printf("Enter the number of elements:");
 scanf("%d",&number);
 printf("\n%d %d",n1,n2);//printing 0 and 1
 for(i=2;i<number;++i)//loop starts from 2 because 0 and 1 are already printed
 {
  n3=n1+n2;
  printf(" %d",n3);
  n1=n2;
  n2=n3;
 }
      getch();
}
```

6. Example of Function with arguments and return value.

```
#include <stdio.h>
#include <conio.h>
int add(int a, int b);        // function prototype

void main()
{
   int n1,n2,sum;

   printf("Enters two numbers: ");
   scanf("%d %d",&n1,&n2);

   sum = add(n1, n2);       // function call
   printf("sum = %d",sum);

   getch();;
}

int add(int a, int b)        // function definition
{
   int result;
   result = a+b;
   return result;               // return statement
}
```

7. Example of Function with arguments and no return value.

```
#include <stdio.h>
#include <conio.h>
void add(int,int);        // function prototype

void main()
{
   int n1,n2,sum;
   printf("Enters two numbers: ");
   scanf("%d %d",&n1,&n2);
   add(n1, n2);       // function call
   getch();
}

void add(int a,int b)        // function definition
{
   int result;
   result = a+b;
   printf("sum = %d",result);
}
```

8. Example of Function with no arguments and no return value.

```
#include <stdio.h>
#include <conio.h>
void add(void);            // function prototype

void main()
{
   int n1,n2,sum;

   printf("Enters two numbers: ");
   scanf("%d %d",&n1,&n2);

   add();                 // function call
   sum=n1+n2;
   printf("sum = %d",sum);

   getch();
}

void add()             // function definition
{
  printf("\nAddition of two numbers is : ");
}
```

9. Example of Function with no arguments and return value

```
#include <stdio.h>
#include <conio.h>
int add(void);        // function prototype

void main()
{
   int sum;
   sum = add();       // function call
   printf("\nsum = %d",sum);
   getch();
}

int add()        // function definition
{
    int n1,n2,result;
    printf("Enters two numbers: ");
    scanf("%d %d",&n1,&n2);
    result = n1+n2;
   return result;                // return statement
}
```

10. Calculate area and circumference of a circle using function.

```
//Calculate area and circumference of a circle using functions
#include<stdio.h>
#include <conio.h>
const float PI = 3.14;
float area(float);
float circum(float);
void main() {
 float radius;
 clrscr();
 printf("Enter radius: ");
 scanf("%f", &radius);
 printf("Area : %.2f\n", area(radius));
 printf("Circumference: %.4f\n", circum(radius));
 getch();
}

float area(float radius)
{
  return PI * radius * radius;
}

float circum(float radius)
{
  return 2 * PI * radius;
}
```

11. C program to find sum of n numbers of using function

```
//C program to find sum of n numbers using function.
#include <stdio.h>
#include <conio.h>
int summation(int);
void main()
{
        int num =0;
        int result =0;
        printf("Enter number ");
    scanf("%d", &num);
    result = summation(num);
        printf("\nsum : %d ",result);
        getch();
}
int summation(int num)
{
        int i;
        int sum =0;
        for(i=1; i<=num; i++)
{
                sum += i;
}
        return sum;
}
```

12. Program to find sum of all digits of number using function.

```
#include <stdio.h>
#include <conio.h>
int sum_of_digit(int);
void main(void)
{
            int n,sum=0;
clrscr();
printf("\Enter any number");
scanf("%d",&n)   ;
sum=sum_of_digit(n);
printf("\nSum of all Digits = %d",sum);
   getch();
}

int sum_of_digit(int no)
{
  int sum=0,digit;
            while(no!=0)
            {
         digit=no%10;
         sum=sum+digit;
         no=no/10;
            }
return(sum);
}
```

13. Program to find largest of two numbers using function.

```
#include <stdio.h>
#include <conio.h>

int max(int, int);

void main ()
{

  int a = 100;
  int b = 200;
  int ret;
  ret = max(a, b);
   printf( "Max value is : %d\n", ret );

  getch();
}


int max(int num1, int num2)
 {
    int result;
   if (num1 > num2)
     result = num1;
  else
     result = num2;
   return result;
}
```

14. Program to find largest number in the given array using function.

```
#include <stdio.h>
#include <conio.h>
int maximum(int a[],int size)
{
int max = a[0];
int i=0;
for(i=1;i<size;i++)
{
if(max < a[i])
max = a[i];
}
return max;
}

void main()
{
int i=0,max=0;
int array[] ={9,52,77,635,889,679,6,54,78,34};

printf ("List of Numbers:\n");
for(i=0;i<10;i++)
{
printf ("%d",array[i]);
printf ("\n");
}

max = maximum(array,10);
printf("\nLargest Number is: %d",max);
getch();
}
```

15. Example of passing a string to function.

```
#include <stdio.h>
#include <conio.h>
void myFunction(char name[]);

void main()
{
  myFunction("John");
  myFunction("Jenny");
  myFunction("David");
  getch();
}
void myFunction(char name[])
{
  printf("Hello %s\n", name);
}
```

16. Example of passing a string and one variable to function.

```
#include <stdio.h>
#include <conio.h>
void myFunction(char name[], int age)
{
  printf("Hello %s. You are %d years old\n", name, age);
}

void main() {
  myFunction("Liam", 3);
  myFunction("Jenny", 14);
  myFunction("Anja", 30);
  getch();
}
```

17. Program to check whether the given number is even or odd using function.

```
#include <stdio.h>
#include <conio.h>
int checkEvenOdd(int);
void main()
{
int no,result;
printf("Enter The Number To Check Even or Odd\n");
scanf("%d", &no);
result=checkEvenOdd(no);
if (result==1)
{
printf("%d is Even\n",no);
}
else
{
printf("%d is Odd\n",no);
}
getch();
}
int checkEvenOdd(int num)
{
if(num%2==0)
return (1);
return (0);
}
```

Programs on pointers.

18. To print value and address of variable.

```
#include <stdio.h>
#include <conio.h>
void main()
{
int i=3;
printf("\nValue of i = %d",i);
printf("\nAddress of i = %u",&i);
getch();
}
```

19. To print value and address of variable.

```
#include <stdio.h>
#include <conio.h>
void main()
{
int i=3;
printf("\nValue of i = %d",i);
printf("\nAddress of i = %u",&i);
printf("\nValue of i = %d",*(&i));
getch();
}
```

20. To print value and address of variable.

```
#include <stdio.h>
#include <conio.h>
void main()
{
int i=3;
int *j;
j=&i;
printf("\nAddress of i = %u",&i);
printf("\nAddress of i = %u",j);
printf("\nAddress of j = %u",&j);
printf("\nValue of j = %u",j);
printf("\nValue of i = %d",i);
printf("\nValue of i = %u",*(&i));
printf("\nValue of i = %d",*j);
getch();
}
```

21. Use of mathematical function pow().

```
#include<conio.h>
#include<stdio.h>
#include <math.h>
void main()
{

  printf("%f", pow(4, 3));

 getch();
}
```

22. Use of mathematical function sqrt().

```
#include <stdio.h>
#include <math.h>

void main()
{
float no;
printf("\nEnter any no to calculate it's Square-Root");
scanf("%f",&no);
  printf("\nSquare-Root of %f = %f",no, sqrt(16));

 getch();
}
```